

Games Engineering

WM9M4-15 — revision saved on 28th May 2026 at 08:31

Deadline: 31st May 2026

Approved on 27th May 2026 at 12:08.

Basic module details

Month of introduction/change

October

Year of introduction/change

2026

Module title

Games Engineering

Module duration

4 weeks

Department for SITS

WMG

Summary of changes or other notes for reviewers

The module is technical and a 100% assessment suits delivery across 4 weeks. Positive module reviews and PTES feedback suggest this approach is working well, so no need to change. Agreed with Course Leader and AQ.

Rationale for introduction, revision or discontinuation

This is module 3 in the MSc Games Engineering course. It follows from PFA and Computer Graphics. It can be run in conjunction (as in - with the same prerequisites as) with module 4 Advanced Computer Graphics. It will cover software development aspects as required by games engineering.

Module leader

Kurt Debattista (0673615)

Level

7

Level for SITS

M1

Credit value

15

Module delivery

Blended

Study locations

Location	Primary
University of Warwick Main Campus, Coventry	Yes

Module description

Introductory description

Video game development is one of the most complex software engineering processes, requiring development in a number of wide ranging areas, including but not limited to computer graphics, physics, acoustics, AI, and networking. These typically take the form of sub-systems which can be programmed independently but with a firm understanding of the required specifications and always with sound engineering principles and high performance in mind. This module will provide students with the required set of tools to be able to develop all such subsystems and the right set of engineering practices to be able to integrate them into a coherent whole.

Principal module aims

This module aims to provide students with the fundamental algorithmic, mathematical and programming skills not related to computer graphics that are required to develop high-end video games. In particular it will focus on the software skills required to integrate many gaming sub systems and optimising these systems.

Learning outcomes

- Demonstrate a comprehensive and critical understanding of the components and systems that underpin contemporary video game development
- Design and develop multiple subsystems used in games
- Efficiently integrate multiple sub systems into a games engine/application

Outline syllabus for publication

Games loop

Games software development - patterns etc.

AI for games - Pathfinding, Behaviour trees,

Acoustics

UI

Multithreading

Optimisations

Events/messaging

Game Databases

Game networking

Current C++ standards as required in industry

Should this module have a reading list?

Yes

Talis Aspire link

<https://rl.talis.com/3/warwick/lists/1760E120-7AB5-1772-D788-2F4456959053.html>

Indicative reading list

Nystrom, R. (2014). Game programming patterns. Genever Benning.

Gamma, E., Helm, R., Johnson, R., Johnson, R. E., & Vlissides, J. (1995). Design patterns: elements of reusable object-oriented software. Pearson Deutschland GmbH.

Herlihy, M., Shavit, N., Luchangco, V., & Spear, M. (2020). The art of multiprocessor programming. Newnes.

Module delivery

Minimum student number

20

Maximum student number

87

Estimated student number

30

Teaching split

Department	Percentage
WMG	100%

Skills, competencies, and attributes

Subject specific skills

Mathematical skills and programming skills.

Transferable skills

Technology literacy, adaptability.

Includes research element

No

Interdisciplinary

Yes

Interdisciplinary description

The skills developed here can find application in a number of different fields in computing such as AI, networking, data bases, programming embedded systems etc.

International

No

Learning hours breakdown

Lecture

Required: 15 × 1 hour Optional: 0 × 0 hours

Seminar

Required: 0 × 0 hours Optional: 0 × 0 hours

Tutorial

Required: 15 × 1 hour Optional: 0 × 0 hours

Project Supervision

Required: 0 × 0 hours Optional: 0 × 0 hours

Demonstration

Required: 0 × 0 hours Optional: 0 × 0 hours

Practical class

Required: 0 × 0 hours Optional: 0 × 0 hours

Supervised practical class

Required: 0 × 0 hours Optional: 0 × 0 hours

Fieldwork

Required: 0 × 0 hours Optional: 0 × 0 hours

External visit

Required: 0 × 0 hours Optional: 0 × 0 hours

Work based learning

Required: 0 × 0 hours Optional: 0 × 0 hours

Online learning (scheduled sessions)

Required: 0 × 0 hours Optional: 0 × 0 hours

Online learning (independent)

Required: 5 × 1 hour Optional: 0 × 0 hours

Autonomous skills development (Degree apprenticeship provision only)

Required: 0 × 0 hours Optional: 0 × 0 hours

Other activity

Required: 20 × 1 hour Optional: 0 × 0 hours

Details of other activity

20 hours of student self directed study in preparation for in class work. Guidance on self directed study will be provided in class.

Placement

0 hours

Private study

35 hours

Details of private study and independent learning

Further reading around the state-of-the-art in game systems.

Assessment

60 hours

Total study hours for module

150 hours

Assessment

Reassessment model

Individual

Assessment components

Title	Type	Length	Weight	Final chronological	Eligible for self-certification
Games Engineering Project	Project - Individual	4000 words	100%	Yes	Yes (extension)
Student preparation and completion time (hours)	60				
Description	Write a system that uses multiple aspects of games engineering, such as multithreading and socket programming.				
Reassessment component					
Title	Type	Length		Eligible for self-certification	
Games Engineering Project	Project - Individual	4000 words		No	
Description	Develop a system that incorporates multiple aspects of games engineering covered on the module, for example parallel computing and networking.				

Must the student pass all parts of the assessment to pass the module?

Yes

Methods for providing feedback on assessment

Written feedback.

Assessment learning outcomes

	Games Engineering Project (100%)
Demonstrate a comprehensive and critical understanding of the components and systems that underpin contemporary video game development	X
Design and develop multiple subsystems used in games	X
Efficiently integrate multiple sub systems into a games engine/application	X

Availability

Availability of module

Course	Route	Year of study	Type of availability	Required
MSc in Games Engineering		1	Core	Yes

Has pre-requisite modules?

Yes

Pre-requisite modules

- All of
WM9M2-15 Computer Graphics

Has post-requisite modules?

No

Has anti-requisite modules?

No

Any additional resource requirements

n/a